

0. Data Needed for Each SmartLoad Module

SmartLoad needs **different types of datasets** because each module learns something different. Below is a simplified explanation of the data required for forecasting, anomaly detection, and reinforcement learning (RL).

0.1 Forecasting (Prediction)

For forecasting, we need **long, continuous time-series data of system usage**. This usually includes hourly readings.

This data trains forecasting models that predict future load (such as the next hour or next day), helping SmartLoad prepare resources in advance.

0.2 Anomaly Detection

Anomaly detection needs data that contains both **normal patterns and abnormal behavior** (faults or unusual spikes). Models learn how to detect sudden changes, instability, or unexpected values.

Examples:

- Numenta Anomaly Benchmark (NAB) — labeled normal and abnormal time series.
- Yahoo SMD — multivariate server metrics with injected anomalies.

This helps SmartLoad identify unhealthy or failing nodes early.

0.3 Reinforcement Learning (RL)

RL **does not rely on a fixed dataset**. Instead, it needs a training environment that **simulates real-world behaviour**. The environment provides:

- states (current load, latency, resource usage),
- actions (routing, scaling decisions),
- rewards (reduced latency, lower cost).

RL agents interact with the environment, learning how to make better routing and scaling decisions over time.

1. Dataset Summary Table

Dataset / Tool	URL / Source	Format	Open	SmartLoad Module(s)
Google Borg Cluster Traces	https://github.com/google/cluster-data	CSV / protobuf	Yes (CC-BY)	RL, Forecasting
Alibaba Microservices Traces (2021)	https://github.com/alibaba/clusterdata	CSV	Yes	RL, Forecasting, Anomaly Detection
Microsoft Azure Public Dataset	https://github.com/Azure/AzurePublicDataset	CSV (VM metrics)	Yes	Forecasting
BitBrains Business Workloads (2013)	Research corpus	CSV (VM metrics)	Yes	Forecasting
PlanetLab VM Traces (2011)	Public trace by Beloglazov	CSV (CPU usage)	Yes	Forecasting
Numenta Anomaly Benchmark (NAB)	https://github.com/numenta/NAB	Time-series CSV	Yes (MIT)	Anomaly Detection
Yahoo Server Machine Dataset (SMD)	https://github.com/NetManAI0ps/OmniAnomaly	Multivariate CSV	Yes	Anomaly Detection
Mininet Network Emulator	http://mininet.org/	Synthetic data	Yes (BSD)	RL, Anomaly Detection
CloudSim Simulator	https://github.com/Cloudslab/cloudsim	Simulated workloads	Yes	RL, Forecasting
Synthetic Anomaly Injector	Custom scripts	Synthetic time series	Yes	Anomaly Detection

2. Data Selection and Integration

SmartLoad selects datasets that contain metrics identical or similar to those collected by its telemetry pipeline. Examples include:

- Google Borg traces: per-job CPU/memory usage and task execution patterns.
- Alibaba microservice traces: call graphs, response times, and service-level latencies.
- Azure, BitBrains datasets: historical VM utilization logs for forecasting training.
- PlanetLab CPU traces: 5-minute sampled VM load, suitable for time-series modeling.

These datasets are integrated into the Telemetry Database for model training and benchmarking. Historical traces help train forecasting models (ARIMA, LSTM, Prophet), while RL training uses real workloads to pretrain policies.

3. Reinforcement Learning (RL) Data Usage

RL training relies on decision-epoch data such as:

- node load and utilization,
- queue depth,
- task arrival rates,
- observed latency and throughput.

Google Borg and Alibaba traces provide realistic clustered workloads, enabling RL agents to learn routing strategies. Mininet can generate synthetic latency/loss patterns for RL training when real network telemetry is unavailable.

4. Anomaly Detection Data Usage

SmartLoad's anomaly detection module requires datasets with labeled failures or injected anomalies. The following datasets support this:

- NAB: real and artificial time series with labeled anomalies.
- Yahoo SMD: multivariate server metrics with injected anomalies.

For missing failure types, synthetic anomaly injection scripts generate additional training signals.

5. Forecasting and Autoscaling Data Usage

Forecasting models require long historical sequences of CPU, memory, or request-rate data. Datasets used include:

- Azure VM metrics,
- BitBrains workloads,
- PlanetLab CPU traces.

These datasets help SmartLoad predict:

- load spikes,
- resource saturation,
- off-peak intervals,
- autoscaling requirements.

6. Pipeline Connections

Datasets feed into the SmartLoad telemetry pipeline as follows:

1. The telemetry DB stores both real-time and historical metrics.
2. The decision layer uses the data for RL, forecasting, and anomaly detection.
3. Autoscaler uses forecasting predictions for proactive scaling.

Simulation tools (CloudSim, Mininet) fill gaps when real datasets lack certain metrics (queue depth, network delays).

7. Compatibility and Limitations

- Google/Alibaba traces are large and require downsampling.
- Some datasets lack specific metrics (queue lengths, network RTT).
- Preprocessing is required: timestamp alignment, normalization, and feature extraction.

- Synthetic data is used for missing failure modes.

8. References (Datasets)

All datasets mentioned are publicly available:

- Google Borg: <https://github.com/google/cluster-data>
- Alibaba Cluster Traces: <https://github.com/alibaba/clusterdata>
- Numenta NAB: <https://github.com/numenta/NAB>
- Yahoo SMD: <https://github.com/NetManAIOps/OmniAnomaly>
- CloudSim: <https://github.com/Cloudslab/cloudsim>
- Mininet: <http://mininet.org/>